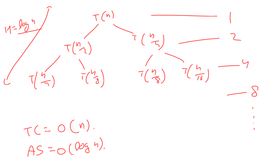
$T(n) = T(\frac{n}{2}) + T(\frac{n}{2}) + c.$



$TC = O(n).$
$AS = O(\log n).$

"hello" ✗

abcba ✓

abccba ✓

TC = O(n)
AS = O(1)

levels

race a car. ✗

↓

race a car.

↓

race a car.

↓

A man; Nama ✓

↓

a man; nama

↓

amanamama

[ch >= 'A' && ch <= 'Z'] → Uppercase check

[ch >= 'a' && ch <= 'z'] → Lowercase check

[ch >= '0' && ch <= '9'] → Digit check

[Uppercase to lowercase]
ch = 'A' + 'a'

[Lower to Upper]
ch = 'a' + 'A'

ch = 'D'
'D' − 'A' + 'a' = 'd'
3

A man; Nama
n n

TC = O(n).
AS = O(1)

isSame.

a = b ← Same
a-'A'+'a' = = b ← U
b-'A'+'a' = = a ← L

a

b

Same
L.
U

---

a^b

res = 1
i = 0 → b-1
res = res * a.

TC = O(b).
AS = O(1).

i ' a ' a ' a . .
b-times

$a^b = a^{b/2} \times a^{b/2}$. if b is even.
$= a^{b/2} \times a^{b/2} \times a$. if b is odd

TC = O(log b).
AS = O(log b)

---

n = 12343 1
n/10
1234 3
n/10
123 4
n/10
12 3
n/10
1 2
n/10
0

cnt = 0 1 2 3 4
5 6

TC = O(log n)
AS = O(1).